

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA14

COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO3: *Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

1. A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server **scanme.nmap.org**. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using **Nmap**, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on **scanme.nmap.org**. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.

Objective

To perform reconnaissance of scanme.nmap.org and collect information about its network services using Nmap.

Target: scanme.nmap.org

Tool: Nmap

i. Host Check

Command 1 : nmap -sn scanme.nmap.org

Result:

The host was detected as active.

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$
kali@kali:~$ nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:56 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00061s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Nmap done: 1 IP address (1 host up) scanned in 8.34 seconds

kali@kali:~$ nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:57 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0011s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 886 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
5060/tcp  closed isakmp
5061/tcp  closed isakmp
993/tcp   closed imaps
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap

Nmap done: 1 IP address (1 host up) scanned in 104.31 seconds

kali@kali:~$ nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:59 EDT
Stats: 0:00:05 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 100.00% done; ETC: 06:59 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0055s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http        Apache httpd 2.4.7 ((Ubuntu))
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8081/tcp  closed http-proxy
```

ii. TCP Enumeration

Command 2 : `nmap -sS scanme.nmap.org`

Result:

The scan displayed the TCP ports and their current states.

```
Session Actions Edit View Help
5061/tcp open  sip-tls?
8080/tcp closed http-proxy
8081/tcp closed blackice-icecap

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 98.98 seconds

kali@kali:~$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:01 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0008s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 986 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap
8085/tcp  closed unknown
OS fingerprint not ideal because: maxTimingRatio (3.404000e+00) is greater than 1.4
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 88.24 seconds

kali@kali:~$ nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:03 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.018s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 23.26 seconds

kali@kali:~$
```

Port Details:

TCP Port	Service	Status
22	SSH	Open
80	HTTP	Open

iii. Identify Running Services , Identify OS, Default NSE Scripts

Command 3 : nmap -sV scanme.nmap.org

Command 4 : sudo nmap -O scanme.nmap.org

Command 5 : nmap -sC scanme.nmap.org

Result:

- Nmap determined the service information associated with the discovered ports.
- OS fingerprinting was performed to determine the probable operating-system family.
- Nmap's default scripts collected additional service information.

Security Observations

The enumeration showed which services are exposed to the Internet. This information can be useful to an attacker for reconnaissance, vulnerability research, and identifying possible authentication attack targets.

Security Measures

- Remove services that are not required.
- Restrict remote administration access.
- Apply security patches regularly.
- Configure a firewall using least privilege.
- Monitor suspicious connection attempts.

Final Result

Nmap provided useful information about the target's network exposure, including reachable services, ports, service information, and OS characteristics. The assessment demonstrates the importance of minimizing and securing publicly exposed services.

2. The security team has deployed a deliberately vulnerable Linux server, **192.168.56.101 (Metasploitable 2)**, in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use **Nmap** to perform comprehensive enumeration of **192.168.56.101**. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.

Objective :

To perform comprehensive enumeration of the Metasploitable 2 Linux server using Nmap and identify its open ports, running services, service versions, operating system, and potential security risks.

Target : Metasploitable 2

IP Address : 192.168.56.102

Network : Host-Only Adapter

Tool : Nmap

Procedure

- Installed Metasploitable 2 in VirtualBox.
- Connected Kali and Metasploitable using a Host-Only network.
- Verified the target IP as 192.168.56.102.
- Tested connectivity using : `ping -c 4 192.168.56.102`
- Confirmed the host was active using : `nmap -sn 192.168.56.102`
- Performed Nmap enumeration to identify ports, services, versions, OS information, and NSE results.

Commands and Results

A. Host Discovery

Command : `nmap -sn 192.168.56.102`

Result : The target was found to be active.

B. TCP Port Scanning

Command : `sudo nmap -sS -p- 192.168.56.102`

Result : Nmap identified the open TCP ports on the target.

```

kali@kali:~$ cat /dev/null > /tmp/.zsh_history
kali@kali:~$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=3.79 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.72 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=3.82 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.31 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 303ms
rtt min/avg/max/mdev = 1.723/2.710/3.789/0.773 ms

kali@kali:~$ nmap -sn 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 08:20 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0017s latency).
NMC Address: 08:00:27:22:8a:10 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds

kali@kali:~$ sudo nmap -sS -p- 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 08:21 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0011s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  cproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
4432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u

```

C. Service Version Detection

Command : `sudo nmap -sV 192.168.56.102`

Result : The services and their versions were identified.

```

Pentest ESX
Scanner the WorldWide Web

--kali@kali:~$
--kali@kali:~$ nmap -vv 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2020-06-18 00:21 EDT
Nmap scan for 192.168.56.102
Host is up (4.000ms latency).
Not shown: 977 closed tcp ports (reset)
Host: 192.168.56.102
21/tcp open ftp vsftpd 2.3.4
22/tcp open ssh OpenSSH 4.7p1 Debian 6ubuntu1 (protocol 2.0)
23/tcp open telnet Linux telnetd
25/tcp open smtp Postfix smtpd
53/tcp open domain ISC BIND 9.4.3
80/tcp open http Apache/2.2.2.2 (Ubuntu) DAV/2
113/tcp open rshd rshd 2 (RSC 3100001)
139/tcp open netbios-ssn Samba smbd 3.0 - 4.0 (workgroup: WORKGROUP)
443/tcp open netbios-ssn Samba smbd 3.0 - 4.0 (workgroup: WORKGROUP)
312/tcp open ssn netkit-ssn strand
313/tcp open login OpenSSH or Solaris rlogind
334/tcp open shell Netkit rsh
4444/tcp open java-rmi dno Classpath: gmxregistry
1024/tcp open bindshell Metasploitable root shell
4444/tcp open rfx 2.4 (CPC 3100001)
1122/tcp open ftp ProFTPD 1.3.1
1580/tcp open mysql MySQL 5.4.51a-Debian
5432/tcp open postgresql PostgreSQL 9.4.3 - 9.4.3
5900/tcp open vnc VNC (protocol 3.3)
6000/tcp open x11 (screen denied)
6067/tcp open irc UnrealIRCd
6070/tcp open k9rb Apache/2.0.6 (Debian) v1.3
6106/tcp open http Apache/2.0.6 (Debian) 2SP engine 1.1
MAC Address: 08:00:27:71:76:10 (PCS SystemTechnica/Oracle VirtualBox virtual NIC)
Service Info: hosts: metasploitable.localdomain, inc.Metasploitable.smb; cbs: smb, linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 13.27 seconds

--kali@kali:~$
--kali@kali:~$ nmap -vv 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2020-06-18 00:22 EDT
Nmap scan report for 192.168.56.102
Host is up (4.000ms latency).
Not shown: 977 closed tcp ports (reset)
Host: 192.168.56.102
21/tcp open ftp vsftpd 2.3.4
22/tcp open ssh OpenSSH 4.7p1 Debian 6ubuntu1 (protocol 2.0)
23/tcp open telnet Linux telnetd
25/tcp open smtp Postfix smtpd
53/tcp open domain ISC BIND 9.4.3
80/tcp open http Apache/2.2.2.2 (Ubuntu) DAV/2
113/tcp open rshd rshd 2 (RSC 3100001)
139/tcp open netbios-ssn Samba smbd 3.0 - 4.0 (workgroup: WORKGROUP)
443/tcp open netbios-ssn Samba smbd 3.0 - 4.0 (workgroup: WORKGROUP)
312/tcp open ssn netkit-ssn strand
313/tcp open login OpenSSH or Solaris rlogind
334/tcp open shell Netkit rsh
4444/tcp open java-rmi dno Classpath: gmxregistry
1024/tcp open bindshell Metasploitable root shell
4444/tcp open rfx 2.4 (CPC 3100001)
1122/tcp open ftp ProFTPD 1.3.1
1580/tcp open mysql MySQL 5.4.51a-Debian
5432/tcp open postgresql PostgreSQL 9.4.3 - 9.4.3
5900/tcp open vnc VNC (protocol 3.3)
6000/tcp open x11 (screen denied)
6067/tcp open irc UnrealIRCd
6070/tcp open k9rb Apache/2.0.6 (Debian) v1.3
6106/tcp open http Apache/2.0.6 (Debian) 2SP engine 1.1
MAC Address: 08:00:27:71:76:10 (PCS SystemTechnica/Oracle VirtualBox virtual NIC)
Service Info: hosts: metasploitable.localdomain, inc.Metasploitable.smb; cbs: smb, linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 13.27 seconds

```

```
kali@kali -  
Session Actions Edit View Help  
| 100000 2 | 111/udp | rpcbind  
| 100003 2,3,4 | 2049/tcp | nfs  
| 100003 2,3,4 | 2049/udp | nfs  
| 100005 1,2,3 | 424287/udp | mountd  
| 100005 1,2,3 | 484999/tcp | mountd  
| 100021 1,3,4 | 39232/tcp | nlockmgr  
| 100021 1,3,4 | 59638/udp | nlockmgr  
| 100024 1 | 44123/udp | status  
| 100024 1 | 55383/tcp | status  
| 139/tcp open netbios-ssn  
| 445/tcp open microsoft-ds  
| 512/tcp open exec  
| 513/tcp open login  
| 514/tcp open shell  
| 1099/tcp open rmiregistry  
| 1324/tcp open ingreslock  
| 2049/tcp open nfs  
| 2121/tcp open cpcproxy-ftp  
| 3306/tcp open mysql  
| mysql-info:  
| Protocol: 10  
| Version: 5.0.51a-JubuntuS  
| Thread ID: 8  
| Capabilities flags: 43564  
| Some capabilities: SupportsCompression, ConnectWithDatabase, Speaks41ProtocolNew, SwitchToSSLAfterHandshake, SupportsTransactions, Support41Auth, LongColumnFlag  
| Status: Autocommit  
| Salt: '_10{-_GN-p9Hfa6}gm/  
| 5432/tcp open postgresql  
| sql-erts Subject: commonName=jubuntu004-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX  
| Not valid before: 2010-03-17T14:07:45  
| Not valid after: 2010-04-10T14:07:45  
| sql-date: 2020-08-10T12:23:31+00:00; -is from scanner time.  
| 5900/tcp open vnc  
| vnc-info:  
| Protocol version: 3.3  
| Security types:  
| _ VNC Authentication (2)  
| 8009/tcp open x11  
| 6657/tcp open irc  
| 8009/tcp open sjpeg  
| _sjpeg-methods: failed to get a valid response for the OPTIOn request  
| 8180/tcp open unknown  
| _http-title: Apache Tomcat/5.5  
| _http-favicon: Apache Tomcat  
| NIC Address: 08:00:27:73:F8:10 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
  
Host script results:  
| _clock-skew: mean: 59m59s, deviation: 2h00m00s, median: -1s  
| smb-security-mode:  
| account_used: guest  
| authentication_level: user  
| challenge_response: supported
```

```

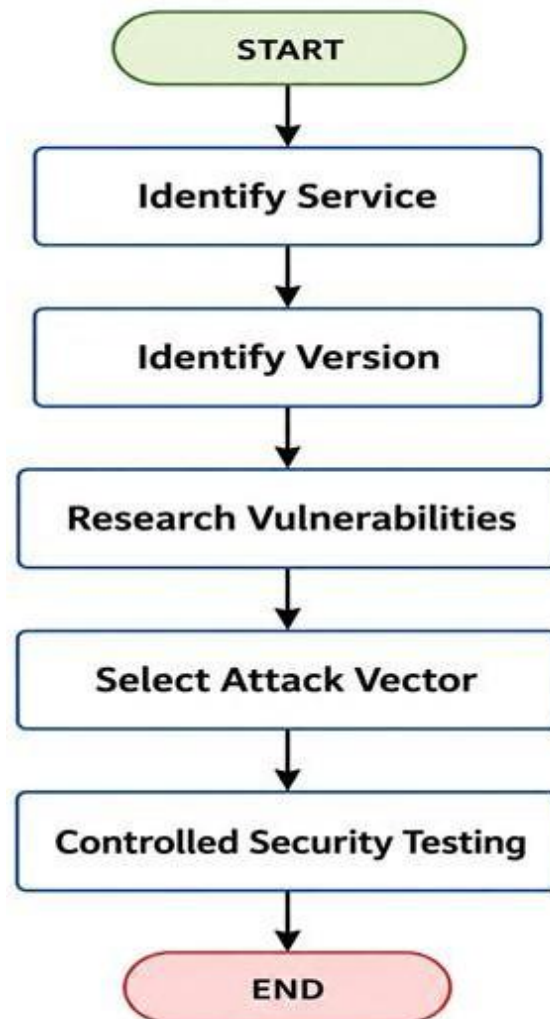
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:73:fb:10
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe73:fb10/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:57 errors:0 dropped:0 overruns:0 frame:0
          TX packets:72 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:8714 (8.5 KB)  TX bytes:6464 (6.3 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$

```

Security Attack Analysis :



Service Risk Analysis :

- FTP : File transfer - High risk
- Telnet : Remote login - High risk
- HTTP : Web service - High risk
- Database : Data storage/access - High risk
- SSH : Secure remote administration - Medium risk
- SMTP : Mail communication - Medium risk
- DNS : Name resolution - Medium risk

Recommendations

- Disable unnecessary services.
- Replace Telnet with SSH.
- Restrict database services from network access.
- Keep all software updated.
- Use strong authentication.
- Configure firewall rules.

Conclusion

Nmap successfully identified the active host, open ports, running services, service versions, and operating-system information of the Metasploitable 2 server. The discovered services represent different levels of security risk and provide potential entry points that should be prioritized during authorized penetration testing.

3. A financial organization wants to assess the security posture of a Windows workstation located at **192.168.56.102** before connecting it to the production network. Using **Nmap**, enumerate **192.168.56.102** to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks

Objective

The objective was to assess the security of the Windows workstation 192.168.56.102 before connecting it to the production network. Nmap was used to identify open ports, services, file-sharing services, remote-management services, and OS details.

Target: 192.168.56.102

Tool: Nmap

i. Commands :

Scanning	Commands
Host Availability Scan	ping -c 4 192.168.56.102
TCP Port Scan	sudo nmap -Pn -sV -sC -p 5060,5061 192.168.56.102
Service Version Scan	sudo nmap -Pn -sV -sC -p 5060,5061 192.168.56.102
Windows Services Scan	sudo nmap -Pn -sS -sV -p 135,139,445,3389,5985,5986,22 192.168.56.102
OS Detection Scan	sudo nmap -Pn -O --osscan-guess 192.168.56.102
UDP Service Scan	sudo nmap -Pn -sU -sV -p 5060,5061 192.168.56.102

```
kali@kali: ~  
Session Actions Edit View Help  
Host is up.  
  
PORT      STATE SERVICE VERSION  
22/tcp    filtered ssh  
135/tcp   filtered msrpc  
139/tcp   filtered netbios-ssn  
445/tcp   filtered microsoft-ds  
389/tcp   filtered ms-adt-server  
5985/tcp  filtered wsdman  
5986/tcp  filtered wsdman  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 5.52 seconds  
  
kali@kali:~$ sudo nmap -Pn -O --osscan-guess 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:09 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0061s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter[bridge]general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox:cpus/admny_gosporovsk:slirp:cpus/qemu:qemu  
Aggressive OS guesses: AT&T BGM21a voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 81.47 seconds  
  
kali@kali:~$ sudo nmap -Pn -sU -sV -p 5060,5061 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:11 -0400  
Nmap scan report for 192.168.56.102  
Host is up.  
  
PORT      STATE SERVICE VERSION  
5060/udp  open|filtered sip  
5061/udp  open|filtered sip-tls  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 105.51 seconds  
  
kali@kali:~$
```

```
kali@kali: ~  
Session Actions Edit View Help  
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 0.08% done  
  
kali@kali:~$ sudo nmap -Pn -sS --top-ports 1000 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:03 -0400  
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 3.50% done; ETC: 08:06 (0:03:41 remaining)  
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.00% done; ETC: 08:06 (0:03:36 remaining)  
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.50% done; ETC: 08:06 (0:03:32 remaining)  
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 5.00% done; ETC: 08:06 (0:03:29 remaining)  
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 5.50% done; ETC: 08:06 (0:03:26 remaining)  
Nmap scan report for 192.168.56.102  
Host is up (0.0086s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 69.45 seconds  
  
kali@kali:~$ sudo nmap -Pn -sV -sC -p 5060,5061 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:04 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.024s latency).  
  
PORT      STATE SERVICE VERSION  
5060/tcp  open  sip  
5061/tcp  open  sip-tls7  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 147.42 seconds  
  
kali@kali:~$ sudo nmap -Pn -sS -sV -p 135,139,445,389,5985,5986,22 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:07 -0400  
Nmap scan report for 192.168.56.102  
Host is up.  
  
PORT      STATE SERVICE VERSION  
22/tcp    filtered ssh  
135/tcp   filtered msrpc
```

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
  
kali@kali:~$ ping -c 4 192.168.56.102  
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.  
ping: Warning: invalid tv_usec -88005735319229642 us  
ping: Warning: time of day goes back (-679674803233 s), taking countermeasures  
ping: Warning: invalid tv_usec -88005735319229382 us  
ping: Warning: time of day goes back (-679674803233 s), taking countermeasures  
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms  
wrong data byte #16 should be 0x18 but was 0xc0  
#16  c0 a8 38 66 8 0 d1 93 4a 23 0 3 c9 bc 79 6a 0 0 0 0 d8 5b 2 0 0 0 0 10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319229283 us  
ping: Warning: time of day goes back (-679674803233 s), taking countermeasures  
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x18 but was 0xc0  
#16  c0 a8 38 66 8 0 d1 93 4a 23 0 3 c9 bc 79 6a 0 0 0 0 d8 5b 2 0 0 0 0 10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319229278 us  
ping: Warning: time of day goes back (-679674803233 s), taking countermeasures  
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x18 but was 0xc0  
#16  c0 a8 38 66 8 0 d1 93 4a 23 0 3 c9 bc 79 6a 0 0 0 0 d8 5b 2 0 0 0 0 10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
  
--- 192.168.56.102 ping statistics ---  
4 packets transmitted, 1 received, +2 duplicates, 75% packet loss, time 304ms  
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4  
  
kali@kali:~$ sudo nmap -Pn -sS -p- 192.168.56.102  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 07:58 -0400  
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 0.08% done  
  
kali@kali:~$ sudo nmap -Pn -sS --top-ports 1000 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:03 -0400  
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 3.50% done; ETC: 08:06 (0:03:41 remaining)  
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.00% done; ETC: 08:06 (0:03:36 remaining)  
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.50% done; ETC: 08:06 (0:03:32 remaining)  
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 5.00% done; ETC: 08:06 (0:03:29 remaining)
```

ii. Security Analysis

- Ports 5060 and 5061 are the main exposed services. If SIP/VoIP is not required on this workstation, these services unnecessarily increase the attack surface.
- The SMB, RDP, WinRM, SSH, and NetBIOS ports were filtered, reducing direct exposure to common file-sharing and remote-management attacks.
- An attacker could use exposed SIP services for network reconnaissance and, if the service is vulnerable or poorly configured, potentially attempt unauthorized access or further attacks.

iii. Recommendations

- Disable SIP services if they are not required.
- If required, allow access only from authorized systems.
- Keep SIP software updated and securely configured.
- Maintain firewall restrictions on SMB, RDP, WinRM, and other unnecessary services.
- Disable unnecessary Windows services.
- Use network segmentation and strong authentication.
- Perform another Nmap scan after hardening.

iv. Conclusion

The workstation has a limited exposed attack surface, with SIP-related ports 5060 and 5061 being the main open services. SMB, RDP, WinRM, SSH, and NetBIOS were filtered. Unnecessary SIP services should be disabled or restricted to follow the principle of least privilege and improve the workstation's security before connecting it to the production network.

4. An organization hosts a web application on **192.168.56.103 (OWASP Juice Shop Server)** and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using **Nmap**, perform a detailed enumeration of **192.168.56.103**, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure.

Objective

The objective of this assessment is to enumerate the OWASP Juice Shop server (192.168.56.103) using Nmap and identify its network exposure, open ports, running services, service versions, operating system, and web services.

Tools Used

Kali Linux , Nmap , OWASP Juice Shop

Target IP: 192.168.56.103

Enumeration Procedure

- i. **Host Discovery :** `nmap -sn 192.168.56.103`

Result : This confirms whether the target server is active.

- ii. Port Scanning :** `nmap -p- 192.168.56.103`

Result : This scans all 65,535 TCP ports and identifies accessible services.

```
kali@kali:~$ sudo nmap -sS 192.168.56.103
Nmap scan report for 192.168.56.103
Host is up (0.0000000 latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
8080/tcp   filtered pop3

Service detection performed. Please report any incorrect results at http://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds
```

```
kali@kali:~$ sudo nmap -sT 192.168.56.103
Nmap scan report for 192.168.56.103
Host is up (0.0000000 latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
8080/tcp   filtered pop3

Service detection performed. Please report any incorrect results at http://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 2.50 seconds
```

```
kali@kali:~$ sudo nmap -sV 192.168.56.103
Nmap scan report for 192.168.56.103
Host is up (0.0000000 latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
8080/tcp   filtered pop3

Service detection performed. Please report any incorrect results at http://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 2.50 seconds
```

iii. Service and Version Detection : nmap -sV 192.168.56.103

Result : This identifies the services running on open ports and their versions.

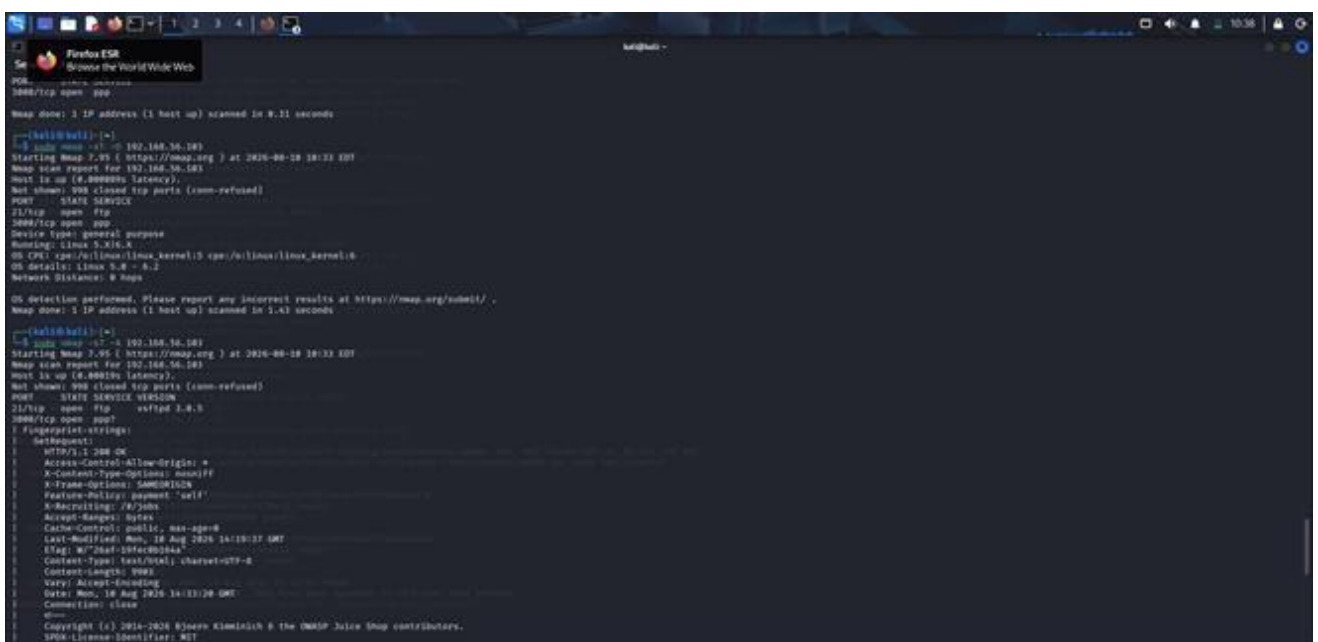
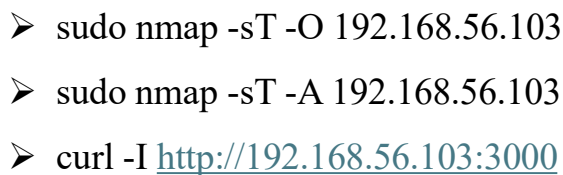
iv. OS Detection : sudo nmap -O 192.168.56.103

Result : This attempts to identify the operating system of the target server.

```
kali@kali: ~  
$ ping -c 4 192.168.56.103  
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:  
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.174 ms  
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.096 ms  
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.063 ms  
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.038 ms  
--- 192.168.56.103 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3062ms  
rtt min/avg/max/mdev = 0.038/0.092/0.174/0.051 ms  
$ nmap -sn 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT  
Nmap scan report for 192.168.56.103  
Host is up.  
Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds  
$ nmap -p- 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT  
Nmap scan report for 192.168.56.103  
Host is up (0.000017s latency).  
Not shown: 65533 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
3000/tcp  filtered ppp  
Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds  
$ nmap -sV 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:24 EDT  
Nmap scan report for 192.168.56.103  
Host is up (0.000050s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 3.0.5  
3000/tcp  filtered ppp  
Service Info: OS: Unix  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds  
$ sudo docker port juice-shop
```

```
kali@kali: ~  
$ ping -c 4 192.168.56.103  
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:  
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.174 ms  
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.096 ms  
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.063 ms  
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.038 ms  
--- 192.168.56.103 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3062ms  
rtt min/avg/max/mdev = 0.038/0.092/0.174/0.051 ms  
$ nmap -sn 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT  
Nmap scan report for 192.168.56.103  
Host is up.  
Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds  
$ nmap -p- 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT  
Nmap scan report for 192.168.56.103  
Host is up (0.000017s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
3000/tcp  filtered ppp  
Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds  
$ nmap -sV 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:24 EDT  
Nmap scan report for 192.168.56.103  
Host is up (0.000050s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 3.0.5  
3000/tcp  filtered ppp  
Service Info: OS: Unix  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds  
$ sudo docker port juice-shop
```

- Result : This identifies HTTP services, page titles, and server information.



Findings

The enumeration provides information about the server's open ports, services, versions, and web application interfaces. The web service is particularly important because it provides direct access to the OWASP Juice Shop application.

Security Analysis

- The discovered services provide an attacker with useful information about the target environment. Service versions can be compared with known vulnerabilities and security advisories.
- Exposed web services can be further assessed for vulnerabilities such as authentication weaknesses, injection, broken access control, cross-site scripting, and security misconfiguration.
- Administrative services such as SSH increase the attack surface if they are unnecessarily exposed or poorly configured.
- Development or supporting services running on non-standard ports should also be reviewed to determine whether they are required.

Recommendations

- Disable unnecessary open ports and services.
- Restrict administrative services such as SSH to trusted networks.
- Keep the operating system and application dependencies patched.
- Use HTTPS with secure TLS configuration.
- Minimize server/version information exposed through HTTP headers.
- Apply firewall rules and network segmentation.
- Regularly scan the server to identify newly exposed services.

Conclusion

Nmap enumeration successfully provides a network-level view of the OWASP Juice Shop server. The information collected helps identify the server's attack surface and provides a foundation for the next phase of penetration testing. The exposed web service should receive particular attention because it directly hosts the application, while unnecessary administrative and supporting services should be restricted to minimize the overall attack surface.

5. The network gateway device at **192.168.56.1** serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using **Nmap**, conduct a comprehensive enumeration of **192.168.56.1**, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

Objective :

To enumerate the gateway device 192.168.56.1 using Nmap, identify exposed ports and services, detect management interfaces, perform OS and script-based enumeration, analyze security risks, and recommend controls to reduce the attack surface.

1. Host Discovery : `sudo nmap -sn 192.168.56.1`

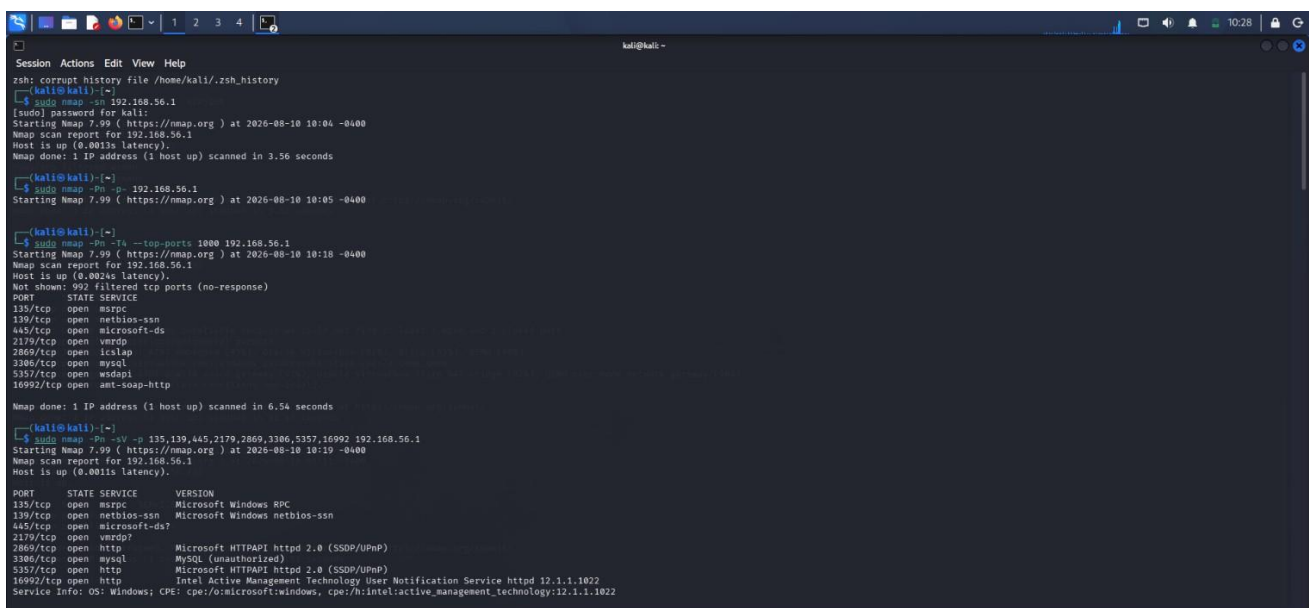
Purpose: Checks whether the target host is active and reachable.

Result: Host 192.168.56.1 is UP.

2. TCP Port Scan : `sudo nmap -Pn -T4 --top-ports 1000 192.168.56.1`

Purpose: Scans the 1,000 most common TCP ports to identify exposed services.

Result: 8 TCP ports were found open: 135, 139, 445, 2179, 2869, 3306, 5357, 16992.



```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ sudo nmap -sn 192.168.56.1
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:04 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0013s latency).
Nmap done: 1 IP address (1 host up) scanned in 3.56 seconds

kali@kali:~$ sudo nmap -Pn -p 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:05 -0400

kali@kali:~$ sudo nmap -Pn -T4 --top-ports 1000 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:18 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0024s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
2179/tcp   open  vmrpd
2869/tcp   open  iclapp
3306/tcp   open  mysql
5357/tcp   open  wsddapi
16992/tcp  open  ant-soap-http

Nmap done: 1 IP address (1 host up) scanned in 6.54 seconds

kali@kali:~$ sudo nmap -Pn -v -p 135,139,445,2179,2869,3306,5357,16992 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:19 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0011s latency).
PORT      STATE SERVICE          VERSION
135/tcp    open  msrpc            Microsoft Windows RPC
139/tcp    open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?   Microsoft Windows [unauthorized]
2179/tcp   open  vmrpd?          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
2869/tcp   open  http            MySQL (unauthorized)
3306/tcp   open  mysql           MySQL (unauthorized)
5357/tcp   open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
16992/tcp  open  http            Intel Active Management Technology User Notification Service httpd 12.1.1.1022
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1022
```

3. Service and Version Detection : `sudo nmap -Pn -sV -p 135,139,445,2179,2869,3306,5357,16992 192.168.56.1`

Purpose: Identifies the services and available version information running on open ports.

4. UDP Port Scan : `sudo nmap -Pn -sU --top-ports 100 192.168.56.1`

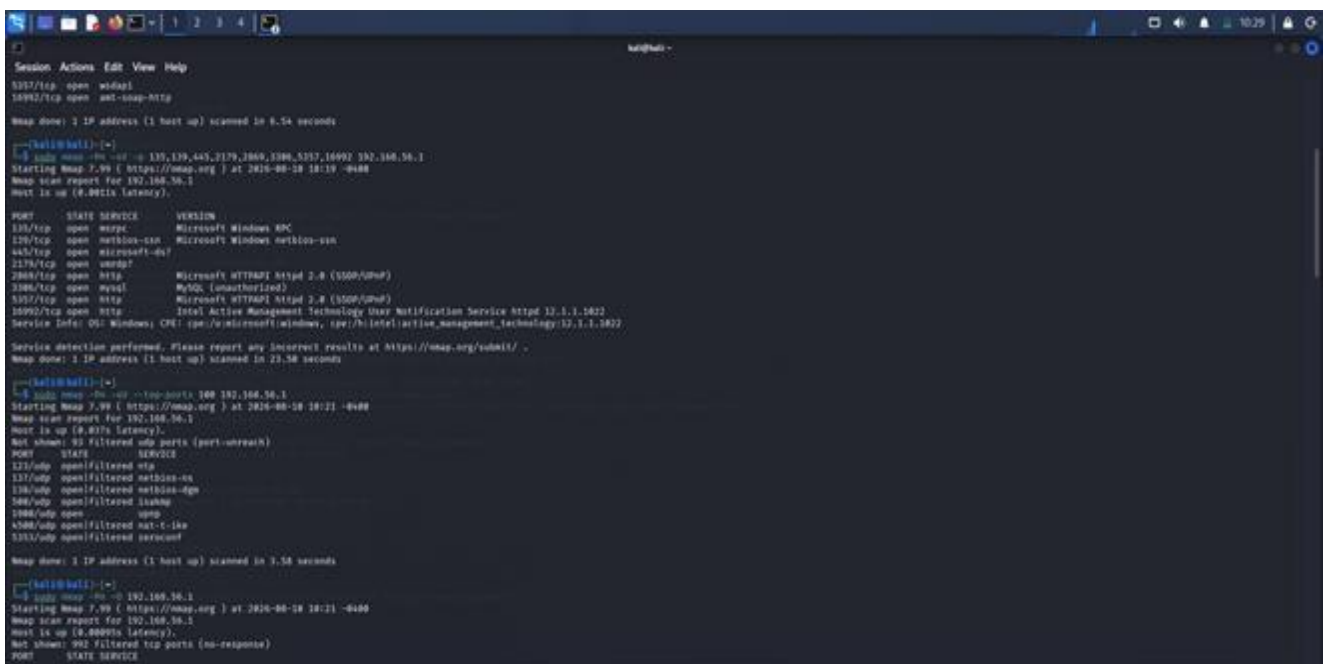
Purpose: Identifies commonly exposed UDP services such as NTP, NetBIOS, IKE and UPnP.

Result: UDP 1900 was confirmed open for UPnP; several other ports were reported as open/filtered.

5. Operating System Detection : `sudo nmap -Pn -O 192.168.56.1`

Purpose: Attempts to identify the operating system and device type.

Result: OS detection was inconclusive. Nmap suggested possible gateway/bridge/virtualized device types, but no exact OS match was confirmed.



```
kali@kali: ~$ sudo nmap -Pn -sV -p 135,139,445,2179,2869,3306,5357,16992 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 18:19 -0400
Nmap scan report for 192.168.56.1
Host is up (0.001s latency).

PORT      STATE SERVICE      VERSION
135/tcp    open  wsdapi
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds
2179/tcp   open  wsdapi
2869/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (500/ups)
3306/tcp   open  mysql        MySQL (unauthenticated)
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (500/ups)
16992/tcp  open  http         Intel Active Management Technology User Notification Service httpd 12.1.1.1802

Service Info: OS: Windows; CPE: cpe:/o:microsoft/windows, cpe:/h:intel/active_management_technology:12.1.1.1802

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 0.54 seconds

kali@kali: ~$ sudo nmap -Pn -sU --top-ports 100 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 18:21 -0400
Nmap scan report for 192.168.56.1
Host is up (0.037s latency).
Not shown: 92 filtered udp ports (port-unreach)

PORT      STATE SERVICE
123/udp    open/filtered ntp
137/udp    open/filtered netbios-ns
138/udp    open/filtered netbios-dgm
146/udp    open/filtered lsknp
1900/udp    open         upnp
4300/udp    open/filtered nat-t-ike
5353/udp    open/filtered mcastmdns

Nmap done: 1 IP address (1 host up) scanned in 1.58 seconds

kali@kali: ~$ sudo nmap -Pn -O 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 18:21 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00000s latency).
Not shown: 992 filtered tcp ports (no-response)

PORT      STATE SERVICE
```

6. Default NSE Script Enumeration : `sudo nmap -Pn -sC -sV -p 135,139,445,2179,2869,3306,5357, 16992 192.168.56.1`

Purpose: Uses Nmap's default scripts along with service detection to collect additional information about exposed services.

Result: SMB security information, HTTP information and Intel AMT information were identified.

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 3.58 seconds  
kali@kali:~$ sudo nmap -Pn -o 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:21 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00095s latency).  
Not shown: 992 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
2179/tcp  open  vmrpd  
2869/tcp  open  iclslap  
3306/tcp  open  mysql  
5357/tcp  open  wsddapi  
16992/tcp open  amt-soap-http  
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny.gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T 60m210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)  
No exact OS matches for host (test conditions non-ideal).  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 10.70 seconds  
kali@kali:~$ sudo nmap -Pn -sC -sV -p 135,139,445,2179,2869,3306,5357,16992 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:22 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0036s latency).  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?   
2179/tcp  open  vmrpd?  
2869/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)  
3306/tcp  open  mysql        MySQL (unauthorized)  
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)  
|_ http-server-header: Microsoft-HTTPAPI/2.0  
|_ http-title: Service Unavailable  
16992/tcp open  http         Intel Active Management Technology User Notification Service httpd 12.1.1.1022  
|_ http-server-header: Intel(R) Active Management Technology 12.1.1.1022  
|_ http-title: IntelSreg; Active Management Technology  
|_ Requested resource was /login.htm  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1022
```

7. Management Port Check : `sudo nmap -Pn -sV -p 22,23,80,443,161,162 192.168.56.1`

Purpose: Checks common management services such as SSH, Telnet, HTTP, HTTPS and SNMP.

Result: All checked management ports were closed.

8. SMB Security Enumeration : `sudo nmap -Pn -p 139,445 --script smb-protocols,smb2-security-mode,smb2-time 192.168.56.1`

Purpose: Identifies SMB versions and checks SMB security settings.

Result: SMB 3.1.1 message signing was enabled and required.

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 8.98 seconds  
kali@kali:~$ sudo nmap -Pn -p 2869,5357 --script http-title,http-headers,http-methods 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:26 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00092s latency).  
PORT      STATE SERVICE  
2869/tcp  open  iclslap  
5357/tcp  open  wsddapi  
Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds  
kali@kali:~$ sudo nmap -Pn -sU -p 1900 --script upnp-info 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:26 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.81s latency).  
PORT      STATE SERVICE  
1900/udp  open  upnp  
|_ upnp-info:  
|   uuid:8651b45e-e605-4600-a665-cb16ea8d3512  
|   location: http://192.168.56.1:2869/upnp/udhisapi.dll?content=uuid:8651b45e-e605-4600-a665-cb16ea8d3512  
|   server: Microsoft-Windows/10.0 UPnP/1.0 UPnP-Device-Host/1.0  
|   webserver: Microsoft-Windows/10.0 UPnP/1.0 UPnP-Device-Host/1.0 Microsoft-HTTPAPI/2.0  
|   deviceType: urn:schemas-upnp-org:device:MediaRenderer:1  
|   friendlyName: swetha  
|   modelNumber: Windows Digital Media Renderer  
|   modelDescription: Digital Media Renderer  
|   manufacturer: Microsoft Corporation  
|   services:  
|   |_ urn:schemas-upnp-org:service:RenderingControl:1  
|   |_ urn:schemas-upnp-org:service:AVTransport:1  
|   |_ urn:schemas-upnp-org:service:ConnectionManager:1  
Nmap done: 1 IP address (1 host up) scanned in 3.84 seconds  
kali@kali:~$ sudo nmap -Pn -p 16992 --script http-title,http-headers,http-methods 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:27 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0013s latency).  
PORT      STATE SERVICE  
16992/tcp open  amt-soap-http  
Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds
```

```
Session Actions Edit View Help
| smb2-security-mode:
| 3.1.1:
| _ Message signing enabled and required
| smb2-time:
| _ date: 2026-08-18T14:24:14
| _ start_date: N/A
| _ clock-skew: 1m86s
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 37.94 seconds

(kali@kali)-[~]
└─$ sudo nmap -Pn -v -o 22,23,80,443,161,162 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 10:24 -0400
Nmap scan report for 192.168.56.1
Host is up (2.8s latency).

PORT      STATE SERVICE VERSION
22/tcp    closed ssh
23/tcp    closed telnet
80/tcp    closed http
161/tcp   closed snmp
162/tcp   closed snmptrap
443/tcp   closed https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.95 seconds

(kali@kali)-[~]
└─$ sudo nmap -Pn -sS -sV -o 139,445 --script smb-protocols,smb2-security-mode,smb2-time 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-18 10:24 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0059s latency).

PORT      STATE SERVICE
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds

Host script results:
| smb2-security-mode:
| 3.1.1:
| _ Message signing enabled and required
| smb2-time:
| _ date: 2026-08-18T14:26:07
| _ start_date: N/A
| smb-protocols:
| _ dialects:
| _ 2.0.2
| _ 2.1
```

Service Risk Analysis

- **135 - MSRPC:** Windows communication service;
- **139 - NetBIOS:** Supports network sharing; can help attackers gather information.
- **445 - SMB:** File-sharing service; exposed SMB may allow unauthorized access.
- **2179 - VMRDP:** Service not fully identified, should be checked , disabled if unused.
- **2869/5357 - HTTPAPI/UPnP:** may expose system details.
- **3306 - MySQL:** Database service; should only be accessible to authorized systems.

Security Implications

The exposed services increase the gateway's attack surface. An attacker with network access could perform service enumeration, authentication attacks, or exploit vulnerable services. SMB, MySQL, UPnP, and Intel AMT are the main services requiring attention.

Recommended Security Controls

- Disable unnecessary services and ports.
- Restrict SMB/NetBIOS to trusted systems.
- Restrict MySQL to authorized hosts.
- Disable UPnP if not required.

Conclusion

The assessment found several exposed TCP and UDP services on 192.168.56.1. SSH, Telnet, HTTP, HTTPS, and SNMP were closed, but SMB, MySQL, UPnP, and Intel AMT were accessible.

Restricting unnecessary services and applying proper firewall, authentication, patching, and network segmentation controls will reduce the gateway's security risk.

Code of Conduct:

I M.Tejaswini (192372285) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Tejaswini

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts	
Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model	
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools	
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results	
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	